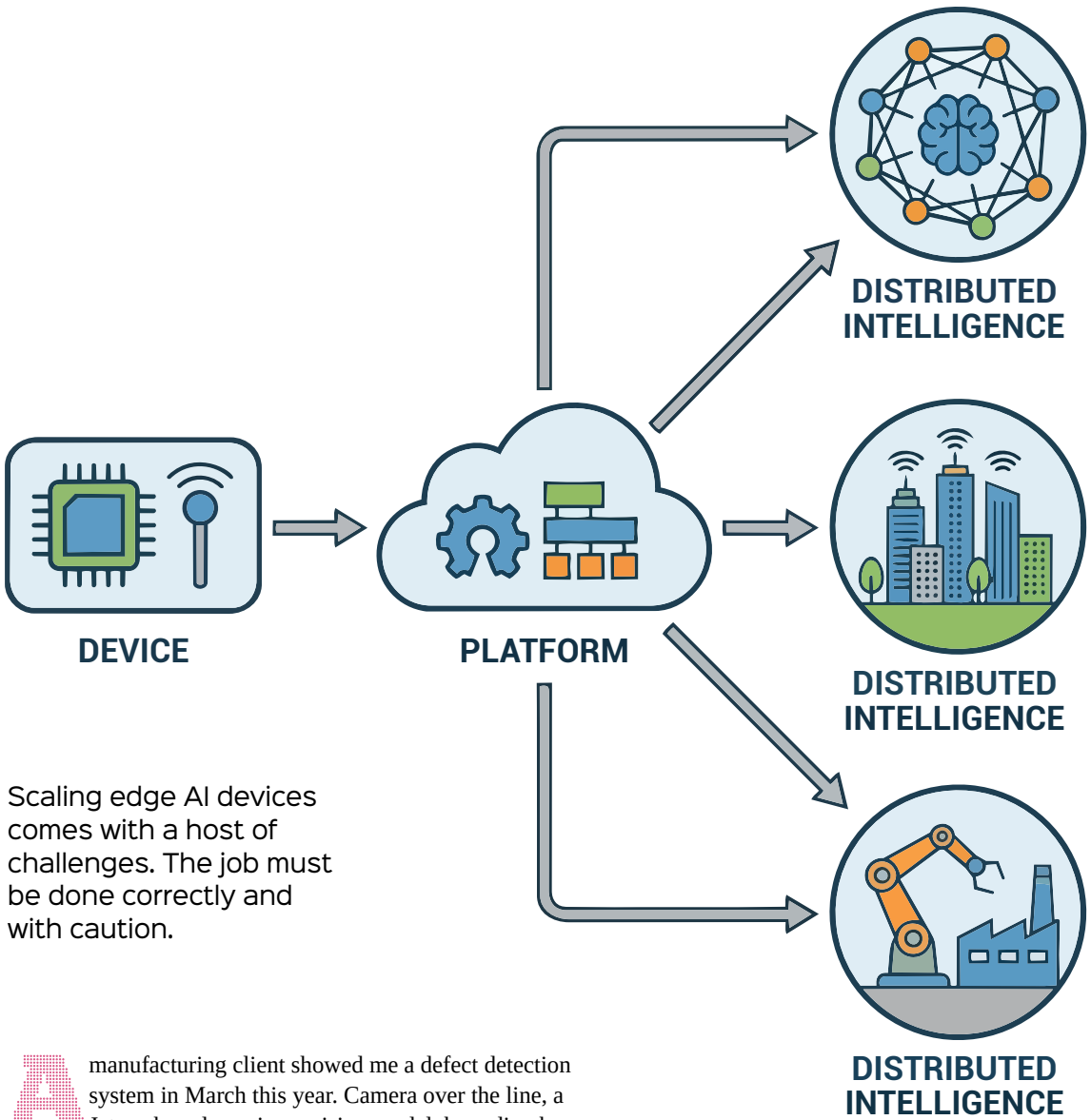
Building an Open Source Edge AI Platform: From Device to Distributed Intelligence



Scaling edge AI devices comes with a host of challenges. The job must be done correctly and with caution.

A manufacturing client showed me a defect detection system in March this year. Camera over the line, a Jetson board running a vision model, bounding boxes drawn in real time. It worked beautifully.

Then they mentioned the plan was forty-two plants!

That is where the conversation changes. One edge AI device is an engineering achievement. Four thousand is a different discipline, and little of what you learned building the first one is of use.

I wrote earlier (*OSFY*, September 2026) that building AI features and operating them are separate problems. At the edge the gap is wider, because you lose both things that make cloud operations tractable: you cannot reach the machine, and the machine is not always online.